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User Perceptions of Trust and Transparency in Music Recommendation Explanations

A Comparison of lyrics-based, Example-Based, and Feature-based explanations

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Attestation of Authorship

We hereby declare that this submission is our work, best upon research that we have conducted.

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Abstract

Music recommender systems increasingly rely on complex algorithms that leave users with limited understanding of why songs are recommended. Explanations have been proposed as a mechanism for improving transparency and trust, yet how different explanation types influence these perceptions remains unclear, and lyrics-based explanations as a user-facing strategy remain underexplored. This study compared three explanation types; lyrics-based, feature-based, and example-based, in terms of their effects on perceived transparency and trust. A within-subjects experimental design was employed, in which 96 participants interacted with a web-based music recommendation prototype and completed validated questionnaires after each condition. Data were analyzed using repeated-measures ANOVA, post-hoc comparisons, interaction analysis, Pearson correlation, and OLS regression. Results showed that explanation type significantly influenced both transparency and trust, but in different ways. Feature-based explanations produced the highest transparency ratings, while lyrics-based explanations generated the highest trust ratings. A significant interaction effect confirmed that transparency and trust represent distinct dimensions of user perception that respond differently to explanation strategies. Moderation analysis further revealed that the transparency–trust relationship was strongest in the lyrics-based condition. The findings suggest that explanation design should balance cognitive clarity with emotional relevance, and that lyrics-based explanations may offer added value for building user trust in music recommender systems.

Keywords: *Music Recommender Systems, User Trust, Transparency, Explainable AI, Explanation Types, Human-Computer Interaction*

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1 Introduction

Music streaming services have become the primary mode of music consumption, offering users immediate access to vast catalogues containing millions of tracks (Velankar & Kulkarni, 2023). While this abundance enables unprecedented access to music, it also creates significant challenges for users who must navigate and make sense of such large collections. Music recommender systems (MRS) play a central role in addressing this challenge by filtering content and supporting music discovery. In industrial settings, these systems are typically built upon complex machine learning models that optimise recommendation accuracy and engagement. However, this growing complexity has raised important concerns regarding users' ability to understand, evaluate, and trust the recommendations they receive.

To address these concerns, explainable recommender systems (XRS) have emerged as a key research area within recommender systems and human AI interaction (Afchar et al., 2022). Explanations are commonly understood as mechanisms that communicate why a particular item was recommended, with the goal of supporting user understanding, transparency, and trust. Prior research has shown that explanations can positively influence users' perceptions of recommender systems, including their perceived usefulness, credibility, and long-term acceptance. In music recommendation specifically, explainability is particularly challenging due to the abstract and subjective nature of musical preferences, as well as the opacity of modern recommendation pipelines.

Existing work highlights that explanations in recommender systems are not a single, uniform concept, but instead encompass a variety of explanation types with different purposes and effects (Afchar et al., 2022). Research distinguishes between explanation forms such as example-based explanations (e.g., past listening behaviour), feature- or content-based explanations (e.g., attributes derived from audio or metadata), and model-oriented or process-oriented explanations that describe how a system operates. Each explanation type emphasises different aspects of the recommendation process and may shape user perceptions in distinct ways, particularly with respect to trust and transparency. For instance, example-based explanations are widely used in commercial music platforms because they are intuitive and easy to communicate, while feature-based explanations can provide more concrete insight into musical characteristics that drive recommendations.

Despite their widespread use, prior studies have identified important limitations in current explanation approaches for music recommender systems (Afchar, 2023; Schedl et al., 2018). Many explanations prioritise simplicity and cognitive efficiency, which can result in shallow or generic justifications that describe correlations rather

than meaningful reasons for a recommendation. Moreover, explanations often reflect system-centric perspectives on how the system processes data rather than user-centric perspectives that align with how listeners interpret and experience music. As a result, explanations may fail to fully support transparency, defined as users' understanding of why a specific recommendation was presented to them in a given context, and may only partially contribute to building user trust.

Parallel to research on explainability, work in music information retrieval and recommendation has increasingly explored the role of lyrics as a rich source of semantic and emotional information (Afchar, 2023; Chepkoech, 2024).

In this context, lyrics-based explanations refer to explanation strategies that justify a recommendation by explicitly referencing the semantic, emotional, or thematic content expressed in a song's lyrics. Example-based explanations communicate relevance by referring to a user's prior listening behaviour (Van Der Waa et al., 2021). feature-based explanations justify recommendations using low-level musical attributes such as tempo, genre, or acoustic features (Deldjoo et al., 2024). Compared to these explanations, Lyrics-based explanations explain song relevance by highlighting shared themes, emotions, and narrative elements present in the lyrics (Gupta et al., 2021).

Such explanations aim to align more closely with how users interpret and relate to music at a meaning-making level. Lyrics convey themes, narratives, and emotions that are often central to how users relate to music, sometimes more directly than audio features or metadata alone. Prior research has demonstrated that incorporating lyrical information can enhance music modelling, emotion recognition, and recommendation relevance. However, this body of work has largely focused on improving recommendation performance rather than on how lyrical information might be leveraged to explain recommendations to users.

While explanations are known to be crucial for trust and transparency in recommender systems, and lyrics are known to be meaningful to users' musical experiences, the intersection of these two research directions remains underexplored (Afchar et al., 2022; Chepkoech, 2024). Existing explanation strategies in music recommender systems primarily rely on example-based or feature-based explanations derived from listening history, audio features, or metadata. The use of lyrics as a basis for user-facing explanations communicating recommendations through lyrical themes, emotions, or semantic content has received little explicit attention in prior work.

This study addresses this gap by investigating lyrics-based explanations as an alternative explanation type in music recommender systems. Lyrics-based explanations are compared specifically with example-based and feature-based explanations because these represent the most established and widely adopted

explanation strategies in existing music recommender systems research and practice (Afchar, 2023; Afchar et al., 2022). Example-based explanations ground recommendations in users' past listening behaviour, while feature-based explanations rely on observable item characteristics such as audio features or metadata. Comparing lyrics-based explanations against these two baselines enables a meaningful evaluation of whether lyrical semantics offer added value beyond established explanation paradigms, rather than assessing lyrics in isolation (Abdul et al., 2018; Damak et al., 2021). By empirically comparing lyrics-based explanations with established example-based and feature-based explanations, the study aims to examine how different explanation types influence key user-oriented outcomes. In particular, the focus is on trust and transparency, as these are central goals of explainable recommender systems and critical factors for sustained user engagement. Through this comparison, the study seeks to contribute to a more nuanced understanding of how explanation design choices shape user perceptions in music recommender systems.

1.1 Problem Statement

Prior research shows that explanations are important for supporting user understanding, transparency, and engagement in music recommender systems. Transparency is a central goal of explanations in recommender systems, as it helps users understand why items are recommended (Afchar, 2023; Afchar et al., 2022). Explainability also plays an important role in fostering user trust, which is essential for sustained user engagement (Afchar et al., 2022). More broadly, adding explanations to recommender systems is said to provide multiple benefits, including increased user trust and system transparency (Wardatzky et al., 2025).

Parallel work in music recommendation has demonstrated that lyrics capture emotional and semantic aspects of music and can enhance recommendation quality and user engagement (Chepkoech, 2024). At the same time, existing explanation approaches in music recommender systems primarily rely on feature-based, example-based, or graph-based explanations derived from audio or interaction data (Afchar, 2023; Afchar et al., 2022; Damak et al., 2021). However, the use of song lyrics as a basis for user-facing explanations in music recommender systems has not been explicitly examined in prior work. This study aims to address this gap by investigating lyrics-based explanations and their effects on user-oriented outcomes such as trust and transparency.

1.2 Purpose and Research Questions

The aim of this study is to explore how different types of explanations in music recommendations affect users' perceptions of trust and transparency in modern music recommender systems. As current streaming platforms increasingly rely on complex and opaque algorithms, users often have limited insight into why particular songs are recommended to them, which makes effective explanations especially important. In this context, the study focuses on whether lyrics-based explanations, which draw on the semantic and emotional meaning of songs, influence users' trust and transparency differently compared to widely used example-based and feature-based explanation approaches. By comparing these explanation types, the study aims to contribute to the design of more understandable, trustworthy, and human-centred music recommender systems that better align with how users experience and interpret music today.

Following from the problem statement, users often lack insight into how and why music recommendations are generated due to the opaque nature of modern recommender systems. Further, it is unclear how different explanation types influence users' perceptions of trust and transparency.

Consequently, the purpose of this study is: To investigate how different explanation types (lyrics-based, example-based, and feature-based) affect users' trust and transparency in music recommender systems. Trust and transparency are central to explainable recommender systems, as they influence both user understanding and confidence in recommendations. Examining how different explanation types impact these two aspects is therefore important.

To fulfil this purpose, the study is guided by the following research question:

RQ1: How do lyrics-based explanations compare with example-based and feature-based explanations in terms of trust and transparency?

1.3 Scope and Delimitations

This study focuses specifically on three explanation types; lyrics-based, feature-based, and example-based, and their effects on perceived trust and transparency in a music recommender system context. The study does not aim to evaluate the technical accuracy or algorithmic quality of the recommendations themselves, as the prototype uses pre-defined song-recommendation pairings rather than a live recommendation engine. The study is therefore concerned with user perceptions of explanations rather than the performance of the underlying system.

The participant sample is limited to individuals familiar with music streaming platforms. The study does not investigate other user groups, such as users unfamiliar with recommender systems, nor does it account for cultural or demographic factors that may influence music preferences or explanation interpretation.

The explanations used in the prototype are manually constructed and designed to be consistent in length and structure across conditions. The study therefore examines explanation type as an isolated variable, rather than explanations generated dynamically by a real recommender algorithm. Generalizability to real-world recommender systems with varying explanation quality or length may therefore be limited.

Finally, this study does not investigate long-term effects of different explanation types on user behavior or trust over time, which represents a direction for further research.

1.4 Outline

Chapter 2 presents the theoretical framework, covering music recommender systems, explainable AI, trust and transparency, and relevant explanation types. Chapter 3 describes the research method, including the study design, artifact creation, data collection instruments, and procedure. Chapter 4 presents the results of the study. Chapter 5 discusses the findings in relation to the research questions and prior literature. Chapter 6 concludes the study and outlines directions for further research.

2 Theoretical Framework

2.1 Music Recommender Systems

Music recommender systems (MRS) are intelligent information filtering systems designed to assist users in discovering relevant music within large digital music collections. With the rapid growth of streaming platforms such as Spotify, Apple Music, and YouTube Music, users now have access to millions of songs, making manual exploration increasingly difficult. Music recommender systems address this challenge by automatically suggesting songs that match users' listening preferences and behaviors (Schedl et al., 2018; Velankar & Kulkarni, 2023).

At a fundamental level, recommender systems aim to predict the likelihood that a user will enjoy a particular item based on historical data and similarity patterns. In the context of music, this involves analyzing both user-related information such as listening history, ratings, playlists, and interaction behavior and song-related information, including audio features, metadata, genre, and lyrical content (Chen & Chen, 2005; Schedl et al., 2022). By processing these data sources, music recommender systems can generate personalized suggestions that align with individual listening preferences.

The importance of music recommender systems has increased significantly in the digital music ecosystem. These systems not only enhance user experience by facilitating music discovery but also influence broader patterns of music consumption. Algorithmic recommendation has become a central mechanism through which listeners encounter new music, shaping listening habits and even affecting music production and distribution practices within the music industry (Shim & Kim, 2026). As a result, recommender systems play an increasingly important role in the cultural and economic dynamics of digital music platforms.

From a technical perspective, music recommender systems rely on complex data-processing pipelines that transform user interactions and music features into personalized recommendations. Hoffmann et al., (2014) describe these systems as multi-layer architectures that integrate data collection, user modeling, similarity computation, and recommendation generation. These components work together to analyze large-scale datasets and identify patterns that indicate potential user preferences.

How Music Recommender Systems Work

The operation of a music recommender system generally involves several key stages.

First, the system collects data from various sources, including user listening histories, playlists, ratings, and contextual information such as time of listening or device type. In addition, information about the music itself is gathered, including metadata (e.g., artist, genre, release year), acoustic features extracted from audio signals, and textual information such as song lyrics (Schedl et al., 2022).

Second, the system constructs a user profile representing the listener's musical preferences. This profile is developed through machine learning algorithms that analyze patterns in past interactions and identify recurring preferences.

Third, similarity calculations are performed to identify relationships between songs or between users. These similarities are then used to predict which songs may be relevant for a specific listener.

Several recommendation techniques are commonly applied in music recommender systems.

Collaborative filtering is one of the most widely used methods. It recommends songs based on similarities between users' listening behaviors. If two users share similar music preferences, songs liked by one user may be recommended to the other (Schedl et al., 2022).

Content-based filtering focuses on the intrinsic characteristics of songs rather than user similarities. These characteristics may include genre, tempo, instrumentation, or acoustic features extracted from the audio signal. Recommendations are generated by identifying songs that share similar attributes with those previously preferred by the user (Deldjoo et al., 2024).

Features of Music Recommender Systems

Modern music recommender systems incorporate several advanced features that enhance user experience and engagement. One important feature is personalized music discovery, which enables users to explore new songs tailored to their individual preferences. These systems also support context-aware recommendations, where factors such as location, time of day, and listening activity are considered when generating suggestions.

Another important feature is the integration of multimodal data sources. In addition to traditional metadata and audio features, recommender systems increasingly incorporate textual data such as lyrics and social signals derived from user interactions. This multimodal approach allows the system to capture richer representations of both songs and user preferences (Schedl et al., 2022).

Furthermore, recommender systems often generate automated playlists and continuous recommendation streams, which dynamically adapt to users' listening behavior over time.

2.2 Explainable Artificial Intelligence in Recommender Systems

Explainable Artificial Intelligence (XAI) refers to methods that make the outputs and decision-making processes of AI systems understandable to humans. According to Abdul et al. (2018), explainability is an essential aspect of human–AI interaction, enabling users to interpret algorithmic decisions and fostering accountability in intelligent systems.

In the context of recommender systems, explanations aim to clarify why certain items are suggested to users. Explanations can help users understand recommendation logic, evaluate the relevance of suggestions, and adjust their preferences accordingly (Afchar et al., 2022). From a user-centered perspective, explanations play a crucial role in improving transparency, trust, and user satisfaction.

Research on explainable recommender systems has expanded significantly in recent years. However, much of this work focuses on technical explainability mechanisms rather than on how different types of explanations influence user perception and decision-making (Wardatzky et al., 2025). Understanding how users interpret and evaluate explanations remains a key challenge for designing effective explainable systems.

2.3 Trust and Transparency in Recommender Systems

Trust and transparency are widely recognized as essential factors influencing the adoption and effectiveness of recommender systems. Because recommendation algorithms often operate in complex and opaque ways, users may find it difficult to understand why particular items are suggested. This lack of understanding can negatively affect users' confidence in the system and reduce their willingness to rely on its recommendations.

Trust in recommender systems refers to the degree to which users believe that the system provides reliable and useful recommendations aligned with their preferences. According to O'Donovan & Smyth (2005), trust plays a crucial role in the interaction between users and recommender systems because users must depend on the system's suggestions without having full knowledge of the underlying algorithms. When users perceive the system as trustworthy, they are more likely to accept recommendations

and continue using the platform. Conversely, when trust is low, users may ignore recommendations or reduce their engagement with the system.

Several factors can influence trust in recommender systems, including perceived recommendation accuracy, system reliability, and the user's understanding of how recommendations are generated. In recent research, explainability has been identified as a key mechanism for improving trust in AI-driven systems. By providing explanations for recommendations, systems can help users interpret algorithmic decisions and better understand why certain items are suggested. Empirical findings by Liu & Wang (2025) indicate that explanations can significantly influence users' trust perceptions, particularly when explanations clearly communicate the reasoning behind recommendations.

Closely related to trust is the concept of transparency. Transparency refers to the degree to which users are able to understand the processes through which a system generates recommendations. In the context of recommender systems, transparency can be achieved by providing explanations that clarify how recommendations are produced. Sinha & Swearingen (2002), emphasize that transparent recommender systems allow users to evaluate the credibility of recommendations and increase users' confidence in the system's decision-making process.

Transparent recommender systems provide users with insight into the logic behind algorithmic decisions, thereby improving users' ability to assess the relevance of recommended items. As a result, transparency can enhance user satisfaction and encourage greater reliance on recommender systems. Furthermore, transparency and trust are closely interconnected, as systems that provide clear explanations are more likely to be perceived as reliable and trustworthy.

In the context of music recommender systems, these aspects are particularly important because music preferences are subjective and influenced by individual tastes and contextual factors. When users receive recommendations that do not clearly align with their expectations, they may question the reliability of the system. Providing understandable explanations can therefore help users interpret recommendations and evaluate their relevance more effectively.

Given the importance of trust and transparency in user interactions with recommender systems, it is essential to investigate how different explanation approaches influence these perceptions. In particular, examining whether alternative explanation formats, such as lyrics-based explanations, can improve users' perceptions of trust and transparency may provide valuable insights for the design of explainable music recommender systems.

2.4 Explanation Types in Recommender Systems

Explanations are a central component of explainable recommender systems, as they help users understand the reasoning behind recommendations and improve user interaction with AI systems. Different explanation strategies have been developed to communicate recommendation logic in ways that are interpretable and meaningful to users. These explanation types vary in terms of the information they present and the mechanisms through which they justify recommendations (Afchar et al., 2022).

In recommender systems research, explanations serve multiple purposes, including increasing user trust, improving transparency, supporting decision-making, and enhancing user satisfaction (Abdul et al., 2018). The effectiveness of explanations depends largely on how the recommendation rationale is communicated to users. Consequently, researchers have explored different explanation formats to determine how they influence user perceptions and interactions with recommender systems.

Among the most commonly studied explanation types are example-based explanations, feature-based explanations, and content-oriented explanations derived from different data sources. In the context of music recommender systems, these explanations can be generated from user behavior, song attributes, or textual information such as lyrics.

Example-Based Explanations

Example-based explanations justify recommendations by referencing items that the user has previously interacted with or shown interest in. In this approach, the system explains a recommendation by highlighting similarities between the recommended item and previously consumed items. For instance, a music recommender system may suggest a song because the user listened to another similar track by the same artist or within the same musical style (Afchar et al., 2022).

Example-based explanations are widely used because they rely on users' prior experiences, making them relatively easy to understand. According to Van Der Waa et al. (2021), example-based explanations provide intuitive reasoning by connecting recommendations to known examples, which can help users quickly interpret why a recommendation was generated. Such explanations leverage familiarity and can strengthen users' perception that the system understands their preferences.

However, example-based explanations may have limitations. Because they rely heavily on historical interactions, they may reinforce existing listening patterns and reduce opportunities for discovering new or diverse content. Additionally, these

explanations may not provide deeper insight into the specific characteristics of recommended items.

Feature-Based Explanations

Feature-based explanations, also referred to as attribute-based or content-based explanations, justify recommendations by highlighting specific characteristics or features of the recommended item. In music recommender systems, these features may include attributes such as genre, tempo, instrumentation, artist similarity, or mood (Afchar et al., 2022).

Content-driven music recommendation approaches rely on the analysis of song attributes to identify similarities between music items (Deldjoo et al., 2024). Feature-based explanations communicate these similarities to users by explicitly describing the attributes that match their preferences. For example, a system may recommend a song because it shares similar acoustic features or genre characteristics with songs the user previously enjoyed.

Feature-based explanations can improve transparency by providing clear information about the characteristics that influence recommendations. By explicitly highlighting relevant attributes, these explanations help users understand the logic behind recommendations and evaluate their relevance (Afchar et al., 2022). However, the effectiveness of feature-based explanations depends on whether the presented features are meaningful and interpretable for users.

Lyrics-Based Explanations

Lyrics represent an important textual component of music that reflects emotional, thematic, and semantic aspects of songs. Unlike traditional recommendation features such as genre or tempo, lyrics provide insight into the narrative and emotional content of music.

Recent research has explored the use of lyrical information for improving music recommendation algorithms. For example, Gupta et al. (2021) proposed a recommendation approach based on semantic similarity between song lyrics, demonstrating that lyrical analysis can help identify meaningful relationships between songs. Similarly, (Chepkoech, 2024) investigated the use of emotional patterns in song lyrics to enhance music recommendations on streaming platforms.

While lyrics have been used as input data for recommendation algorithms, their potential role as explanation mechanisms presented directly to users remains relatively underexplored. Lyrics-based explanations could provide intuitive reasoning

by highlighting thematic or emotional similarities between songs. For instance, a system may explain a recommendation by indicating that a song contains lyrical themes similar to those of songs previously liked by the user.

Because lyrics often reflect emotional and contextual aspects of music, they may provide explanations that are more relatable and meaningful to users compared to purely technical attributes such as audio features. Therefore, investigating lyrics-based explanations may offer new insights into how explanation design influences user perceptions in music recommender systems.

Comparison of Explanation Types

Different explanation types communicate recommendation logic through different forms of information. Example-based explanations rely on users' previous interactions, feature-based explanations highlight attributes of items, and lyrics-based explanations focus on textual and semantic aspects of music.

Each explanation type may influence users' perceptions differently in terms of interpretability, relevance, and credibility. Prior research suggests that explanation formats can significantly affect how users evaluate recommendations and interact with recommender systems (Afchar et al., 2022; Van Der Waa et al., 2021).

Understanding how these explanation types influence key user perceptions such as trust and transparency is therefore important for designing effective explainable recommender systems.

2.5 Technology Acceptance Model (TAM)

This study adopts the Technology Acceptance Model (TAM) as the theoretical framework to examine how different explanation types in music recommender systems influence users' perceptions of transparency and trust. TAM explains how users form attitudes toward technological systems and how these perceptions influence their acceptance and continued use (Davis, 1989).

The model proposes that users' perceptions of a system's characteristics shape their evaluation of the technology and determine whether they consider it reliable, useful, and understandable (Schorr, 2023). In recommender systems, users typically have limited visibility into how recommendations are generated because the algorithms operate in the background (Tintarev & Masthoff, 2015).

As a result, users rely on system cues, such as explanations, to interpret why specific recommendations are presented. These explanations influence how users perceive the

system's behaviour and whether they consider the recommendation process understandable and trustworthy. Within the context of explainable recommender systems, transparency refers to the extent to which users feel they can understand why a recommendation has been made. Explanations are designed to reduce the opacity of algorithmic systems and support user understanding of the recommendation process (Zhang & Chen, 2020).

When users perceive explanations as clear and meaningful, they are more likely to develop confidence in the system's reasoning. In this sense, transparency acts as an important factor influencing trust in automated recommendation systems (Abdul et al., 2018). Trust refers to users' confidence that the recommender system produces reasonable and reliable suggestions that align with their preferences (De Campos et al., 2024). In music recommender systems, trust is particularly important because recommendations involve subjective and emotional experiences. Users may be more willing to rely on recommendations when they perceive the explanation behind them as logical and understandable (De Campos et al., 2024).

In this study, explanation type is treated as the primary system characteristic influencing user perceptions. Three explanation types are examined: example-based explanations, which refer to users' past listening behaviour; feature-based explanations, which rely on musical attributes such as genre or tempo; and lyrics-based explanations, which explain recommendations by highlighting semantic and emotional themes present in song lyrics (Zanon et al., 2025). These explanation strategies provide different types of information that may influence how users interpret and evaluate the recommendation.

Based on the Technology Acceptance Model, explanations that improve users' understanding of the recommendation process are expected to enhance perceived transparency. Higher levels of perceived transparency may in turn contribute to greater trust in the recommender system. By applying TAM in this way, the study investigates how explanation design influences key user perceptions that shape interactions with music recommender systems.

2.6 Research Gap

Although research on explainable recommender systems has grown substantially, several gaps remain in the context of music recommendation. Many existing studies focus primarily on improving algorithmic accuracy rather than examining how explanation design affects user perceptions. In addition, most explanation approaches rely on behavioral similarities or metadata features, while alternative sources such as lyrical information have received limited attention.

Furthermore, there is a lack of research investigating how different explanation types influence key user perceptions such as trust and transparency in music recommender systems. Addressing these gaps is important for the design of recommender systems that are not only accurate but also understandable and trustworthy from a user perspective.

3 Method and Implementation

3.1 Method Choice

This study used an experimental research method to investigate how different explanation types influence users' perceptions of trust and transparency in music recommender systems. An experimental research design is appropriate for this study because it allows the systematic manipulation of an independent variable (explanation type) in order to observe its effect on dependent variables (users' perceived trust and transparency) under controlled conditions. Controlled experiments are a widely used approach in human-computer interaction research for evaluating how changes in system features influence user outcomes (Blandford et al., 2008). This approach has been applied in similar research contexts; for example, (Hamm et al., 2023) used an experimental study to investigate how explainability influences perceived outcomes such as trust and perceived usefulness in explainable AI contexts.

A quantitative approach was chosen to measure participants' perceptions in a structured and comparable manner. This approach facilitates the collection of numeric data through standardized questionnaires, allowing for statistical analysis and objective comparison across explanation conditions (Erdt et al., 2015). While qualitative methods such as interviews could provide deeper insights into user reasoning, they were deemed unnecessary for the current study, as the aim was to systematically compare user perceptions across explanation types rather than to explore individual reasoning in depth.

Alternative research methods, such as between-subject design or mixed-method approaches, were considered but not selected as the primary approach. Although between-subject designs are also commonly used in music recommender system research, this approach requires a large number of participants (Afchar et al., 2022), which is not feasible given the scope and time constraints of this study. Also, mixed-method designs can offer both quantitative and qualitative insights, but would substantially increase study design complexity and data collection time (Mesyitahsyare & Hermina, 2025), and were therefore not chosen.

The experimental, within-subject approach therefore best aligns with the research aim of systematically comparing explanation types and assessing their impact on user perceptions.

Research Design

The study follows a within-subjects design, where each participant experiences all three explanation conditions. This approach is commonly used in music recommender

system research, particularly when participant numbers are limited (Afchar et al., 2022), which makes it suitable for controlled user studies such as this one. This design was selected to reduce variability caused by individual differences (Charness et al., 2012) in musical preferences, listening habits, or degree of prior familiarity with recommender systems. By exposing each participant to all conditions, the design allows for direct comparisons of how different explanation types affect the same individual's perception (Charness et al., 2012) of trust and transparency. Counterbalancing is used to mitigate potential order effects.

The independent variable in this study is the explanation type (example-based, feature-based, and lyrics-based explanations). The dependent variables are users' perceived trust and perceived transparency of music recommendations.

The study focuses on three explanation types commonly used or proposed in music recommender systems:

- Example-based explanations, which justify recommendations by referring to users' past listening behaviour.
- Feature-based explanations, which explain recommendations using observable musical characteristics such as genre, tempo, or mood.
- Lyrics-based explanations, which justify recommendations by highlighting semantic, emotional, or thematic elements present in song lyrics.

These explanation types are treated as distinct conditions and are presented to participants through an interactive prototype. Participants experienced each explanation type while evaluating a set of music recommendations, after which they provide feedback through standardized questionnaires measuring trust and transparency. This setup ensured a controlled comparison of explanation effects across conditions.

Hypotheses

Given the potential differences between explanation types, this study investigated how lyrics-based explanations, example-based explanations, and feature-based explanations influence users' perceptions of music recommender systems.

In particular, the study examined the impact of explanation types on two important user perceptions: trust and transparency. Since explanations can shape how users interpret and evaluate recommendations, different explanation formats may produce varying levels of trust and perceived transparency

H1: Explanation type affects users' perceived transparency of music recommendations.

Different explanation approaches provide varying levels of information about how recommendations are generated. Explanations that clearly communicate the reasoning behind recommendations can improve users' understanding of the system and enhance perceived transparency (Abdul et al., 2018; Sinha & Swearingen, 2002). Therefore, it is expected that the type of explanation provided by a music recommender system will influence how transparent users perceive the recommendation process.

H2: Explanation type affects users' perceived trust of music recommendations.

Explanations not only improve understanding but can also affect users' confidence in algorithmic systems. When users receive clear and meaningful explanations, they are more likely to perceive the system as reliable and trustworthy (Liu & Wang, 2025; O'Donovan & Smyth, 2005). Consequently, different explanation strategies may lead to different levels of trust in the recommender system.

In addition to H1 and H2, a supplementary analysis was conducted to examine the relationship between perceived transparency and perceived trust across explanation types. This analysis is presented at the end of the Results section.

3.2 Artifact Creation

As part of this research, an interactive web-based prototype of a music recommender system was developed to serve as the experimental instrument through which participants experienced the three explanation conditions. The prototype was not intended to represent a fully functional recommender system, but rather to simulate the user-facing experience of receiving music recommendations with different types of explanations in a controlled and realistic manner.

The prototype was developed using HTML, CSS, and JavaScript, and was hosted publicly via GitHub Pages to allow remote access for participants. The interface was designed to resemble a music streaming application, drawing on established design conventions from platforms such as Spotify, in order to provide participants with a familiar and ecologically valid interaction context. The visual design was first conceptualized in Figma before being implemented in code.

The prototype consisted of several sequential screens: an informed consent page, an instruction page, a practice round, a song selection screen, three recommendation screens (one per explanation condition), and a questionnaire screen following each

condition. Participants first selected a song they liked from a curated list of eight well-known tracks. Based on this selection, the system displayed four recommended songs. The same set of recommended songs was shown across all three conditions, with only the accompanying explanation changing between conditions. This design decision ensured that any differences in participant responses could be attributed to the explanation type rather than to variation in the recommended songs themselves.

To control for order effects, participants were randomly assigned to one of six possible orderings of the three explanation conditions at the start of each session. Questionnaire responses, along with a randomly generated anonymous participant identifier, were automatically submitted to a Google Sheets database via a Google Apps Script endpoint upon completion of the study.

3.3 Data Collection

Data for this study were collected through a controlled user interaction with a music recommendation prototype, followed by self-report questionnaires measuring perceived trust and transparency. This approach is well-established in recommender systems research; Tintarev & Masthoff (2007) identify questionnaires as one of the primary methods for measuring explanation goals such as trust and transparency.

Participants and Sampling

An a priori power analysis was conducted using GPower to determine the required sample size. A repeated measures ANOVA (within-subjects design) was selected, as each participant was exposed to all explanation conditions. Assuming a medium effect size ($f = 0.25$), an alpha level of 0.05, a statistical power of 0.95, and three measurements corresponding to the three explanation types, the analysis indicated a required sample size of 43 participants.

A total of 96 participants completed the study, exceeding the required sample size. This larger sample increases the statistical power of the analysis and strengthens the reliability of the findings. Since each participant evaluated all three explanation conditions, the final dataset consisted of 288 condition-level observations for analysis.

Participants were recruited through convenience sampling via social media and messaging platforms. The inclusion criteria required participants to be 18 years or older and familiar with digital music platforms and recommender systems, as the study involves interaction with a music recommendation system prototype. Participants were required to confirm that they were at least 18 years of age before accessing the survey. This criterion was included to ensure that all participants were legal adults and able to provide informed consent for participation. No specialised

technical knowledge was required, as the study focuses on user perceptions captured through interactive tasks and questionnaires rather than algorithmic understanding. Eligibility was self-reported by participants upon providing informed consent. Although convenience sampling may limit the generalizability of findings to wider populations, it enabled efficient recruitment within the time constraints of the thesis project.

Measurement Instruments

Transparency

Perceived transparency was measured using items adapted from the transparency scale proposed by (Hellmann et al., 2022), which was specifically developed to assess users' perception of transparency in recommender systems. The authors note that there are currently few established measurement instruments for evaluating perceived transparency in recommender systems, which motivated the development of this scale.

The scale conceptualizes transparency as a multidimensional construct consisting of input, functionality, output, and interaction dimensions. For the purpose of this study, items related to output and functionality were selected, as these dimensions are most relevant for evaluating explanations accompanying music recommendations.

Minor wording adjustments were made to adapt the items to the music recommendation context. For example, the term "items" was replaced with "songs". Items related to system interaction or underlying data usage were excluded because they were not relevant to the prototype used in this experiment.

1. I understood which song characteristics were considered to generate recommendations.
2. I understood why the songs were recommended to me.
3. I understood why the system determined that the recommended songs would suit me.
4. I can tell how well the recommendations match my preferences
5. The system provided information to understand why the songs were recommended.

All items were rated on a 5-point Likert scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*).

Trust

Perceived trust was measured through self-report using a questionnaire. Afchar et al., argues that trustworthiness as an explanation goal should be measured through specific questions such as *"Do you trust the recommendation?"*. Tintarev & Masthoff

(2007) conceptualize trust as one of seven key goals of explanations in recommender systems, defining it as increasing users' confidence in the system; a framing that guided the operationalization of trust in this study.

The trust items were drawn from two established sources in recommender systems and explainable AI research. First, the item *"The recommender can be trusted"* was included from the ResQue questionnaire (Pu et al., 2011), which is a widely used instrument in RS research where trust is evaluated as part of users' overall attitudes toward the system. Second, three additional items were adapted from the Trust in XAI scale (TXAI) developed by (Hoffman et al., 2023). The TXAI scale was selected as it is specifically designed to measure user trust in the context of explainable AI, making it more appropriate for this study than general automation trust scales such as (Jian et al., 2000). Items were adapted to fit the music recommender system context while preserving the original meaning, and only the best-performing items from the psychometric evaluation were retained. The final items were:

1. I trust the song recommendation based on the explanation.
2. I am confident in the music recommender system based on the explanation provided.
3. The system is very reliable. I can count on it to give good recommendations.
4. I feel safe that when I rely on the system, I will get good song recommendations.

All items were rated on a 5-point Likert scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*).

Procedure

The study was conducted entirely online. Participants accessed the prototype and questionnaire through a single web link, which directed them to a self-guided interactive session requiring no researcher involvement during data collection.

Before beginning the tasks, participants were presented with an informed consent form outlining the purpose of the study, what participation involved, and their right to withdraw at any time. Participants were required to confirm their consent before proceeding. Following consent, a set of instructions was displayed explaining the structure of the study and what participants were expected to do. A short practice round was then provided, allowing participants to familiarize themselves with the prototype interface, the song selection mechanism, and the recommendation screen, before encountering any explanation conditions.

After completing the practice round, participants selected one song they liked from a curated list of eight well-known tracks. This song selection was made once and

remained fixed throughout the entire session, ensuring that all three explanation conditions were evaluated in relation to the same initial input.

Participants were then shown four recommended songs accompanied by explanations. Each participant experienced all three explanation types; lyrics-based, feature-based, and example-based, one at a time, in a randomly assigned order. The same four recommended songs were displayed across all three conditions; only the accompanying explanation changed between conditions. This design decision ensures that any observed differences in participant responses can be attributed to the explanation type rather than variation in the recommended content. Following each condition, participants completed a short questionnaire measuring their perceived trust and transparency. This sequence of recommendations with explanations followed by a questionnaire, was repeated for each of the three conditions.

Upon completing all three conditions and their associated questionnaires, participants were shown a unique anonymous participation code generated at the start of their session. Participants were informed that this code could be used to request deletion of their data if they wished to do so. The session concluded with a thank-you screen confirming that their responses had been recorded.

3.4 Data Analysis

3.4.1 Data Analysis Method

In this study, quantitative statistical analysis was used to examine how different explanation types in music recommender systems influence users' perceptions of transparency and trust. The collected survey data were analyzed using Python, which enabled efficient processing, statistical testing, and visualization of the dataset. The analysis focused on comparing participant responses across three explanation conditions: feature-based explanations, lyrics-based explanations, and example-based explanations.

During the design phase of the research, careful consideration was given to the type of data required to answer the research questions and to the appropriate statistical techniques needed for analysis. Since the study aimed to compare participant perceptions across multiple explanation conditions and examine relationships between transparency and trust, several statistical methods were selected to address different aspects of the research problem.

The analysis began with descriptive statistics, including means, standard deviations, minimum values, maximum values, quartiles, and response distributions. Descriptive analysis was important because it provided an overview of participant responses

within each explanation condition and allowed initial observation of differences between transparency and trust perceptions. Variables such as transparency mean scores and trust mean scores were summarized for all conditions to identify general response patterns.

To statistically examine whether explanation type significantly influenced transparency and trust perceptions, repeated-measures ANOVA tests were conducted. This method was appropriate because the same participants evaluated all explanation conditions, making the data dependent and repeated across conditions (Shani & Gunawardana, 2011). The repeated-measures ANOVA allowed comparison of the mean differences between feature-based, lyrics-based, and example-based explanations while controlling for participant-level variability.

Following the ANOVA analysis, Bonferroni-corrected paired t-tests were conducted as post-hoc analyses. These tests identified which explanation conditions differed significantly from one another for both transparency and trust outcomes. The Bonferroni correction was applied to reduce the risk of Type I error caused by multiple comparisons.

To further investigate whether transparency and trust behaved differently across explanation types, a two-way repeated-measures ANOVA was conducted. This analysis examined the interaction between explanation type and outcome type (trust versus transparency). The interaction analysis was particularly important because it allowed the study to determine whether explanation strategies influenced transparency and trust uniformly or whether the effects differed depending on the user perception being measured.

In addition, Pearson correlation analysis was conducted to examine the relationship between transparency and trust perceptions. Correlation analysis was relevant to the study because the theoretical framework, the Technology Acceptance Model (TAM), suggests that user perceptions such as transparency and trust may be interconnected within technological systems.

Finally, moderation analysis using Ordinary Least Squares (OLS) regression was performed to investigate whether explanation type altered the relationship between transparency and trust. This analysis helped determine whether certain explanation strategies strengthened or weakened the influence of transparency on user trust.

The significance level for all statistical analyses was set at .05, which is commonly used in behavioural and social science research. Results with p-values below .05 were considered statistically significant, while p-values below .001 were interpreted as highly significant.

3.5 Validity and Reliability

Validity and reliability were important considerations throughout the research process to ensure that the findings accurately reflected participant perceptions and that the study maintained methodological credibility.

Construct validity was considered during the development of the survey instrument. The questionnaire items measuring transparency and trust were designed based on concepts commonly discussed in explainable recommender systems and Technology Acceptance Model literature. Transparency-related questions focused on users' understanding of recommendation reasoning, while trust-related questions focused on confidence in the recommender system and perceived reliability of recommendations. By aligning the survey items with existing theoretical concepts, the study aimed to ensure that the questionnaire accurately measured the intended constructs.

Internal validity was addressed by presenting all participants with the same recommendation scenarios and explanation structures. Each participant evaluated all three explanation types under similar conditions, reducing inconsistencies in exposure between conditions. Additionally, standardized survey procedures were followed throughout data collection to minimize variation in participant experience.

External validity may be limited due to the use of convenience sampling and the relatively limited participant population. Since the study focused on a specific survey sample, the findings may not fully represent all music streaming users or broader recommender system populations. Nevertheless, the participant responses provided valuable insight into user perceptions of explainable music recommendation systems.

Reliability refers to the consistency and repeatability of the study's measurements and procedures. The use of structured Likert-scale survey items improved consistency in participant responses across conditions. Furthermore, all statistical analyses were conducted systematically using Python, ensuring reproducibility of the calculations and analytical procedures. By using standardized analytical methods such as repeated-measures ANOVA, correlation analysis, and regression analysis, the study aimed to maintain consistent and reliable interpretation of the collected data.

The study also acknowledges certain limitations related to validity and reliability. Participant responses were self-reported, meaning that perceptions of transparency and trust were based on subjective evaluation rather than direct behavioural observation. Additionally, user familiarity with recommender systems and individual musical preferences may have influenced responses differently across participants. Recognizing these limitations is important for interpreting the findings appropriately and identifying opportunities for future research.

3.6 Considerations

Several considerations were addressed throughout the conduct of this study, including ethical, scientific, and societal considerations related to explainable recommender systems and user data collection.

From an ethical perspective, participants were informed about the purpose of the study before completing the survey. Participation was voluntary, and respondents were allowed to withdraw from the study at any stage. No personally sensitive information was collected from participants, and all survey responses were handled anonymously to maintain participant privacy and confidentiality.

Scientific considerations were particularly important due to the subjective nature of trust and transparency perceptions. Since music experiences are highly personal and emotional, participant responses may vary depending on individual musical preferences, familiarity with recommendation systems, and personal interpretation of explanations. The study therefore focused on identifying overall perception patterns rather than assuming universally consistent user behaviour.

Societal considerations were also relevant because recommender systems increasingly influence how users discover music, media, and digital content. As algorithmic systems become more integrated into everyday digital experiences, understanding how explanation strategies affect user trust and transparency becomes increasingly important. The study contributes to broader discussions surrounding explainable artificial intelligence (XAI), transparency in algorithmic systems, and the design of more human-centred recommendation technologies.

Finally, the study considered the importance of balancing interpretability with user experience. While transparency is often emphasized in explainable AI research, the findings of this study suggest that emotionally meaningful explanations may also influence user trust in important ways. This consideration highlights the complexity of designing recommendation explanations that are both understandable and psychologically meaningful to users.

4 Results

4.1 Descriptive Analysis

Table 4.1 presents the descriptive statistics for transparency and trust across the three explanation conditions, including the mean, standard deviation, minimum, quartile, median, and maximum values. These statistics provide additional insight into both the central tendency and distribution of participant responses.

Statistical Measure	Feature-based		lyrics-based		Example-based	
	Transparency	Trust	Transparency	Trust	Transparency	Trust
Mean	4.49	3.37	4.32	4.22	3.35	3.89
SD	0.43	0.64	0.61	0.76	0.60	0.61
Min	4.0	2.25	2.0	1.50	1.0	2.50
25%	4.0	3.00	4.0	3.94	3.0	3.50
Median	5.0	3.25	4.5	4.38	4.0	4.00
75%	5.0	3.75	5.0	5.00	4.0	4.25
Max	5.0	5.0	5.0	5.0	5.0	5.0
N	96	96	96	96	96	96

Table 4.1: Descriptive statistics for perceived transparency and trust across explanation types. Ratings were measured on a 1–5 Likert scale.

For transparency, feature-based explanations consistently received the highest evaluations across participants. The average transparency rating was 4.49 out of 5, indicating that participants generally perceived feature-based explanations as highly clear and understandable. The median value was 5.0, while the lower and upper quartiles remained high at 4.0 and 5.0, respectively. This shows that most participants consistently rated feature-based explanations positively in terms of transparency. The relatively low standard deviation value of 0.43 further suggests strong agreement among participants. In addition, the minimum transparency score for this condition was 4.0, indicating that even the lowest ratings remained comparatively high.

lyrics-based explanations also produced high transparency evaluations. The average transparency score was 4.32, suggesting that participants generally considered lyrics-based explanations understandable and informative. The median value of 4.5

and upper quartile value of 5.0 indicate that many participants evaluated these explanations positively. However, compared to the feature-based condition, the wider spread of responses and slightly higher standard deviation of 0.61 suggest greater variability in participant perceptions. The minimum value of 2.0 also indicates that a small number of participants perceived lyrics-based explanations as less transparent.

Example-based explanations generated the lowest transparency ratings overall. The mean transparency score was 3.35, indicating that participants perceived these explanations as less effective in helping them understand why recommendations were generated. Although the median value remained relatively moderate at 4.0, the lower quartile dropped to 3.0 and the minimum value reached 1.0. This wider distribution suggests greater disagreement among participants regarding the transparency of example-based explanations.

A different pattern emerged for trust. lyrics-based explanations produced the highest trust evaluations overall. The average trust score was 4.22, indicating that participants generally perceived lyrics-based explanations as highly trustworthy and reliable. The median trust value was 4.38, while the upper quartile reached the maximum scale value of 5.0. These findings suggest that a large proportion of participants responded positively to lyrics-based explanations in terms of trust. Although the minimum value was relatively low at 1.5, the overall distribution remained concentrated toward higher trust ratings.

In contrast, feature-based explanations produced the lowest trust scores despite generating the highest transparency ratings. The average trust score for feature-based explanations was 3.37, with a median value of 3.25. The lower quartile value of 3.0 indicates that many participants evaluated these explanations less positively in terms of trust. This finding suggests that explanations based primarily on technical musical characteristics may improve users' understanding of the recommendation process without necessarily increasing confidence in the recommender system itself.

Example-based explanations produced moderate trust evaluations. The average trust score was 3.89, with a median value of 4.0. Compared to transparency, trust ratings in the example-based condition appeared relatively stronger, suggesting that familiarity-based explanations may support trust more effectively than they support understanding of recommendation logic.

Overall, the descriptive statistics demonstrate that transparency and trust did not follow the same pattern across explanation types. Feature-based explanations consistently promoted stronger perceptions of transparency, whereas lyrics-based explanations generated stronger perceptions of trust. These patterns provide

preliminary evidence that transparency and trust represent distinct user perceptions that may be influenced differently by explanation design.

4.2 Repeated-Measures ANOVA

To investigate whether different explanation types significantly influenced users' perceptions of transparency and trust in music recommender systems, repeated-measures ANOVA tests were conducted. This statistical method was appropriate for the study because the same participants evaluated all three explanation conditions: feature-based, lyrics-based, and example-based explanations. The analysis therefore allowed comparisons between conditions while accounting for repeated participant responses.

Outcome	**F**	**df**	**p**	**η^2**
Transparency	159.29	2, 188	< .001	0.46
Trust	54.71	2, 188	< .001	0.21

Table 4.2: Repeated-measures ANOVA results for transparency and trust across explanation types. η^2 represents effect size (values ≥ .14 indicate a large effect).

The repeated-measures ANOVA is directly relevant to this thesis because the study aims to examine whether different explanation strategies produce significantly different user perceptions of transparency and trust. The F-value represents how large the differences between explanation conditions are relative to random variation in participant responses. In behavioural research, higher F-values generally indicate stronger differences between groups. In this study, both transparency and trust produced relatively high F-values, particularly transparency with F = 159.29, indicating substantial variation between the explanation conditions.

The p-value determines whether the observed differences are statistically significant. In social science research, values below .05 are generally considered statistically significant, while values below .001 indicate extremely strong significance and a very low probability that the findings occurred by chance. In this analysis, both transparency and trust produced p-values below .001, indicating highly significant results.

The η^2 value represents effect size and indicates how much variance in participant responses is explained by explanation type. In behavioural studies, values around .01 are considered small, .06 medium, and .14 or above large. The transparency analysis produced $\eta^2 = 0.46$, while trust produced $\eta^2 = 0.21$. Both values exceed the threshold

for large effects, demonstrating that explanation type had a strong practical influence on user perceptions within the recommender system.

These findings indicate that the type of explanation presented to users significantly influenced both transparency and trust perceptions.

4.3 Post-Hoc Comparisons: Transparency

Since the repeated-measures ANOVA revealed significant differences between explanation conditions for transparency, Bonferroni-corrected paired t-tests were conducted to identify which explanation types differed significantly from one another. This analysis is relevant to the thesis because it allows a more detailed comparison of how individual explanation strategies influenced transparency perceptions.

Comparison	**t**	**p (adjusted)**
Example-based vs Feature-based	−15.99	< .001
Example-based vs lyrics-based	−11.77	< .001
Feature-based vs lyrics-based	3.62	< .001

Table 4.3: Post-hoc comparisons for transparency scores between explanation types using Bonferroni-corrected paired t-tests

The post-hoc analysis revealed statistically significant differences between all explanation conditions regarding perceived transparency. The comparison between example-based and feature-based explanations produced the largest difference, with a t-value of -15.99 and a significance level below .001. The negative t-value indicates that participants rated example-based explanations significantly lower in transparency compared to feature-based explanations.

Similarly, the comparison between example-based and lyrics-based explanations produced a t-value of −11.77 with a significance level below .001, indicating that lyrics-based explanations were also perceived as significantly more transparent than example-based explanations.

The comparison between feature-based and lyrics-based explanations produced a smaller but still statistically significant difference. The analysis generated a t-value of 3.62 with a significance level below .001, showing that feature-based explanations achieved significantly higher transparency ratings than lyrics-based explanations.

Overall, the results demonstrate a clear ordering of transparency perceptions across conditions. Feature-based explanations generated the highest transparency ratings, followed by lyrics-based explanations, while example-based explanations produced the lowest transparency perceptions.

4.4 Post-Hoc Comparisons: Trust

Bonferroni-corrected paired t-tests were also conducted to examine differences in trust perceptions between explanation conditions. This analysis is relevant because the thesis investigates whether different explanation strategies influence user trust in distinct ways.

Comparison	t	p (adjusted)
Example-based vs Feature-based	6.26	< .001
Example-based vs lyrics-based	−5.47	< .001
Feature-based vs lyrics-based	−8.68	< .001

Table 4.4: Post-hoc comparisons for trust scores between explanation types using Bonferroni-corrected paired t-tests.

The post-hoc analysis identified statistically significant differences between all explanation conditions regarding trust perceptions. The comparison between example-based and feature-based explanations produced a t-value of 6.26 with a significance level below .001, indicating that example-based explanations generated significantly higher trust ratings than feature-based explanations.

The comparison between example-based and lyrics-based explanations produced a t-value of −5.47 with a significance level below .001. This result indicates that lyrics-based explanations were perceived as significantly more trustworthy than example-based explanations.

The largest trust difference was observed between feature-based and lyrics-based explanations. The analysis produced a t-value of −8.68 with a significance level below .001, demonstrating that lyrics-based explanations substantially outperformed feature-based explanations in terms of perceived trustworthiness.

Taken together, these findings show that participants evaluated the trustworthiness of the recommender system differently depending on the explanation strategy used. lyrics-based explanations consistently generated the highest trust perceptions, whereas

feature-based explanations produced the lowest trust ratings despite achieving the strongest transparency performance.

4.5 Two-Way Repeated-Measures ANOVA (Interaction Analysis)

To examine whether explanation types influenced transparency and trust differently, a two-way repeated-measures ANOVA was conducted. This analysis is particularly important for the thesis because the study not only investigates whether explanation strategies affect user perceptions, but also whether they affect transparency and trust in different ways.

Source	**F**	**df**	**p**	**η^2**
Explanation type	60.25	2, 188	< .001	0.16
Outcome type	43.17	1, 94	< .001	0.03
Explanation type × Outcome type	157.46	2, 188	< .001	0.24

Table 4.5: Two-way repeated-measures ANOVA results for explanation type and outcome type (trust vs transparency). η^2 represents effect size (values ≥ .14 indicate a large effect).

The analysis identified a statistically significant main effect of explanation type. The results produced an F-value of 60.25 with a significance level below .001, indicating that participant responses differed substantially across the three explanation conditions regardless of whether transparency or trust was measured. The effect size value of 0.16 represents a large practical effect, demonstrating that explanation type strongly influenced overall user perceptions.

A statistically significant main effect was also observed for outcome type. The analysis produced an F-value of 43.17 with a significance level below .001, indicating that participants evaluated transparency and trust differently overall across the recommender system conditions.

Most importantly, the interaction effect between explanation type and outcome type was highly significant. The interaction analysis produced an F-value of 157.46 with a significance level below .001. The corresponding effect size of $\eta^2 = 0.24$ indicates a large interaction effect, showing that a considerable proportion of variation in participant responses was explained by the combined relationship between explanation type and outcome type.

This interaction effect demonstrates that explanation strategies did not influence transparency and trust uniformly. Instead, the impact of explanation type depended on whether participants evaluated transparency or trust. In particular, feature-based explanations produced stronger transparency perceptions, whereas lyrics-based explanations produced stronger trust perceptions. These findings indicate that transparency and trust respond differently to explanation design within music recommender systems.

Overall, the interaction analysis provides strong evidence that explanation effectiveness is multidimensional rather than uniform. Different explanation strategies appear to support different psychological aspects of user evaluation, reinforcing the distinction between cognitive understanding and perceived trustworthiness in explainable music recommender systems.

4.6 Correlation Analysis

Pearson correlation analysis was conducted to examine the relationship between transparency and trust perceptions. This analysis is relevant to the thesis because the Technology Acceptance Model suggests that user perceptions such as transparency and trust are conceptually related within technological systems.

Variables	**r**	**95% CI**	**p**	**N**
Transparency × Trust	0.24	[0.12, 0.34]	< .001	288

Table 4.6: Overall Pearson correlation between transparency and trust across all conditions (N = 288 condition-level observations). r values range from −1 to 1.

The analysis identified a positive correlation between transparency and trust, with a correlation coefficient of 0.24. This indicates a weak-to-moderate positive relationship, suggesting that higher transparency perceptions were generally associated with higher trust perceptions across the recommendation conditions.

The relationship was statistically significant with a significance level below .001. In addition, the confidence interval ranged from 0.12 to 0.34 and did not include zero, indicating that the positive relationship was reliable within the sample.

Although the relationship was statistically significant, the relatively modest correlation coefficient suggests that transparency alone does not fully explain trust perceptions within music recommender systems. This indicates that additional factors related to explanation design may influence how users develop trust in recommendation systems.

4.7 Correlation Analysis by Explanation Type

Separate correlation analyses were conducted for each explanation condition to examine whether the relationship between transparency and trust differed across explanation strategies. This analysis is relevant because the study investigates whether certain explanation types strengthen or weaken the relationship between transparency and trust.

Condition	r	95% CI	p	Interpretation
lyrics-based	0.65	[0.51, 0.75]	< .001	Strong positive
Example-based	0.51	[0.34, 0.64]	< .001	Moderate positive
Feature-based	0.20	[−0.00, 0.38]	.054	Weak, not significant

Table 4.7: Pearson correlations between transparency and trust within each explanation type.

The results revealed substantial differences in the relationship between transparency and trust across explanation conditions. The lyrics-based condition produced the strongest relationship, with a correlation coefficient of 0.65, indicating a strong positive association between transparency and trust. This suggests that, within lyrics-based explanations, increases in perceived transparency were strongly associated with higher levels of trust in the recommender system.

The example-based condition also demonstrated a statistically significant positive relationship between transparency and trust, although the strength of the relationship was lower compared to the lyrics-based condition. The correlation coefficient of 0.51 indicates a moderate positive association between the two variables.

In contrast, the feature-based condition produced a comparatively weak relationship between transparency and trust. The correlation coefficient was 0.20, and the significance level exceeded the conventional .05 threshold. This indicates that transparency perceptions within the feature-based condition were not significantly associated with trust perceptions.

These findings suggest that the relationship between transparency and trust is not consistent across explanation strategies. In particular, lyrics-based explanations appear to strengthen the connection between understanding and trust, whereas feature-based explanations improve transparency without producing a similarly strong increase in trust. This pattern provides further evidence that transparency and trust represent distinct but conditionally related user perceptions within explainable music

recommender systems. For additional supporting analysis, the correlation heatmaps are provided in Appendix C.

4.8 Moderation Analysis

An Ordinary Least Squares (OLS) regression moderation analysis was conducted to examine whether explanation type influenced the relationship between transparency and trust. This analysis is particularly relevant because the study investigates whether different explanation strategies strengthen or weaken the extent to which transparency contributes to trust perceptions within music recommender systems.

Predictor	**β**	**SE**	**t**	**p**
Intercept	2.17	0.34	6.44	< .001
Transparency	0.51	0.10	5.16	< .001
Feature-based vs Example-based	−0.14	0.72	−0.20	.845
lyrics-based vs Example-based	−1.45	0.55	−2.65	.009
Transparency × Feature-based	−0.21	0.17	−1.24	.216
Transparency × lyrics-based	+0.30	0.14	2.11	.036

Table 4.8: OLS regression analysis predicting trust from transparency and explanation type. Model fit: $F(5,282) = 40.25$, $p < .001$, $R^2 = 0.42$.

The regression model was statistically significant overall, producing an F-value of 40.25 with a significance level below .001. The model explained approximately 42% of the variance in trust perceptions, indicating strong explanatory power for behavioural research. This suggests that transparency and explanation type jointly contributed substantially to predicting user trust in the recommender system.

Transparency itself emerged as a significant positive predictor of trust. The regression coefficient of 0.51 indicates that increases in perceived transparency were associated with higher trust perceptions overall. This finding supports the theoretical assumption that users who better understand recommendation processes are generally more likely to trust the system.

However, the moderation analysis revealed that this relationship differed depending on the explanation strategy presented. The interaction term between transparency and lyrics-based explanations was statistically significant, with a positive regression coefficient of 0.30 and a significance level of .036. This indicates that the positive

relationship between transparency and trust became significantly stronger within the lyrics-based condition.

In practical terms, this means that when lyrics-based explanations increased users' understanding of the recommendation process, this understanding translated more strongly into trust compared to the reference condition. Transparency therefore appeared to be more effective at generating trust when explanations were semantically and emotionally meaningful.

In contrast, the interaction involving feature-based explanations was not statistically significant. This suggests that although feature-based explanations improved transparency perceptions, increased transparency did not significantly strengthen trust within this condition. In other words, users may have understood the recommendation logic more clearly, but this understanding alone was insufficient to substantially increase trust in the recommender system.

These findings provide important evidence that explanation type moderates the transparency–trust relationship. More specifically, lyrics-based explanations appear to strengthen the connection between understanding and trust, whereas feature-based explanations primarily improve cognitive understanding without producing a comparable increase in trust. This result reinforces the broader finding of the study that transparency and trust represent related but distinct dimensions of user evaluation in explainable music recommender systems.

5 Discussion

5.1 Results Discussion

The purpose of this study was to investigate how different explanation types influence users' perceptions of trust and transparency in music recommender systems. More specifically, the study compared feature-based, lyrics-based, and example-based explanations to examine whether different explanation strategies shape user perceptions differently. The findings demonstrated that explanation type significantly influenced both transparency and trust perceptions. However, the influence was not uniform across the two constructs.

Overall, feature-based explanations generated the highest transparency scores, while lyrics-based explanations generated the highest trust scores. Example-based explanations generally performed weaker across both dimensions. Furthermore, the interaction analysis revealed that explanation type affected trust and transparency differently, indicating that the two constructs do not respond identically to recommendation explanations.

Based on the findings of this study and the relationship with previous research, the following discussion is presented.

5.1.1 Transparency and Feature-Based Explanations

The repeated-measures ANOVA revealed a significant effect of explanation type on perceived transparency. Participants rated feature-based explanations as the most transparent explanation strategy among the three conditions. These explanations relied on musical attributes such as genre, tempo, and mood, which may have provided users with clearer and more structured reasoning behind recommendations.

The findings align with previous research emphasizing the role of explanations in improving user understanding and system transparency. According to (Cramer et al., 2008), providing explanations in recommender systems increased users' understanding and acceptance of recommendations within content-based systems. Similarly, (Lu et al., 2019) argued that good explanations should support users' understanding of algorithmic decisions and improve transparency in AI systems.

These findings also align with (Zeng et al., 2024), who identified transparency and interpretability as major challenges within modern music recommender systems. The authors emphasized that recommendation transparency helps users understand how recommendations are generated and how their data is used within recommendation algorithms.

Although lyrics-based explanations also produced relatively high transparency scores, they did not exceed feature-based explanations. One possible explanation is that feature-based explanations present concrete musical attributes that users can evaluate more rationally, while lyrical and emotional explanations may involve more subjective interpretation.

Example-based explanations produced the lowest transparency ratings. This suggests that explanations relying primarily on previous listening behavior may not provide sufficient reasoning for users to fully understand why songs are recommended.

Overall, the findings suggest that structured and attribute-based explanations contribute more effectively to users' cognitive understanding of recommendation systems.

5.1.2 Trust and lyrics-based Explanations

While feature-based explanations produced the highest transparency ratings, lyrics-based explanations generated the highest trust scores. This finding indicates that explanation type plays an important role in shaping user trust in music recommender systems.

These findings support previous research linking explainability to user trust and satisfaction. According to (Zhou et al., 2025), explainable recommendation strategies play an important role in establishing trustworthiness and improving user satisfaction in recommender systems. Similarly, (Ma & Zheng, 2025) explained that explainable AI mechanisms improve user confidence and trust within AI-driven music systems.

The present findings demonstrate that users did not evaluate all explanation strategies equally. lyrics-based explanations generated significantly higher trust ratings than both feature-based and example-based explanations. This suggests that the information provided through lyrics-based explanations was more effective in supporting trust perceptions within the recommender system.

These findings align with (Chepkoech, 2024), who highlighted that lyrics-based recommendation approaches can support user engagement, transparency, and trust in music recommendation systems. The study emphasized that explanation content can influence how users evaluate recommendation outcomes and the recommendation process itself.

In contrast, feature-based explanations produced relatively lower trust scores despite generating the highest transparency ratings. This finding suggests that understanding how a recommendation works does not necessarily guarantee confidence in the

recommendation itself. Previous research by (Millecamp et al., 2019) also showed that explanations influence user confidence and perceived understanding differently depending on users' personal characteristics and cognitive preferences.

Overall, the findings indicate that explanation strategies differ in their ability to support trust perceptions within music recommender systems.

5.1.3 Divergence Between Trust and Transparency

One of the most important findings of this study was the significant interaction effect identified through the two-way repeated-measures ANOVA. The interaction demonstrated that explanation type influenced trust and transparency differently across conditions.

Feature-based explanations maximized transparency but not trust, whereas lyrics-based explanations maximized trust while producing slightly lower transparency scores. These contrasting patterns indicate that trust and transparency represent distinct dimensions of user evaluation rather than interchangeable outcomes.

The findings extend previous explainable recommender system research by showing that transparency and trust do not respond identically to explanation strategies. Although both constructs were significantly influenced by explanation type, the explanation strategy that produced the highest transparency ratings did not produce the highest trust ratings. This suggests that transparency and trust represent distinct dimensions of user evaluation.

These findings align with (Millecamp et al., 2020), who demonstrated that explanations influence several dimensions of user experience differently, including trust, understanding, novelty, and decision support. Similarly, (Shin, 2021) showed that explanations must not only appear understandable but also reliable and meaningful in order to maintain users' trust in AI systems.

The present study contributes to existing literature by empirically demonstrating that explanation strategies may optimize different user outcomes. Feature-based explanations produced the highest transparency ratings, whereas lyrics-based explanations produced the highest trust ratings. These findings indicate that explanation strategies can influence different aspects of user evaluation in different ways.

Overall, the interaction effect suggests that transparency and trust should not be treated as identical outcomes within explainable recommender systems.

5.1.4 Relationship Between Transparency and Trust

The correlation analysis revealed a positive overall relationship between transparency and trust. However, the strength of the relationship varied substantially across explanation conditions.

The lyrics-based condition showed the strongest positive relationship between transparency and trust, while the feature-based condition showed the weakest relationship. This indicates that the association between transparency and trust differed across explanation strategies.

The moderation analysis further supported this interpretation. The relationship between transparency and trust became significantly stronger within the lyrics-based explanation condition. This indicates that the relationship between transparency and trust was stronger in the lyrics-based condition than in the other explanation conditions.

These findings support the argument presented by (Lu et al., 2019), who explained that good explanations should not only provide information but also help users develop meaningful understanding and confidence in algorithmic decisions.

Previous research by (Shin, 2021) also demonstrated that uncertainty and inconsistency in explanations can negatively influence users' trust in AI systems. Consistent with this perspective, the present findings indicate that transparency alone may not fully explain trust perceptions, as explanation strategies that produced high transparency did not necessarily produce equally high trust.

Overall, the findings suggest that transparency and trust are positively related but remain distinct constructs. Their relationship appears to depend on the explanation strategy used and the way users evaluate recommendation explanations.

5.1.5 Interpretation Through the Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) served as the theoretical foundation for this study and guided the development of the research framework, hypotheses, and data collection instrument. The model was used to conceptualize how a system characteristic, namely explanation type, may influence users' perceptions of a music recommender system.

Within the context of this study, transparency and trust were examined as key user perceptions associated with recommendation explanations. The survey instrument was therefore designed to capture participants' evaluations of these perceptions across different explanation conditions. The comparison of feature-based, lyric-based, and

example-based explanations enabled an assessment of how variations in explanation design influence users' evaluations of the recommender system.

Although the study did not directly measure the original TAM constructs of perceived usefulness and perceived ease of use, TAM provided a useful theoretical lens for examining the relationship between system design characteristics and user perceptions. The framework helped structure the investigation of how explanation strategies influence the way users interpret and evaluate recommendation systems.

5.2 Method Discussion

This study adopted a quantitative within-subject experimental design to investigate how different explanation strategies influence users' perceptions of trust and transparency in music recommender systems. Data was collected through an online survey in which participants evaluated feature-based, lyrics-based, and example-based recommendation explanations.

The within-subject design allowed participants to evaluate all explanation conditions, improving the ability to compare user perceptions directly across explanation types while reducing individual variability between participants. The repeated-measures approach also strengthened the statistical analysis by allowing comparisons within the same participant group.

To improve validity and reliability, standardized explanation formats and identical recommendation contexts were used across all conditions. This ensured that differences in user perception were primarily related to explanation type rather than unrelated presentation differences.

The study utilized repeated-measures ANOVA, post-hoc comparisons, correlation analysis, moderation analysis, and interaction analysis to investigate both direct effects and relationships between variables. The interaction analysis provided additional insight into how trust and transparency behave differently depending on explanation strategy.

However, several limitations should be acknowledged. First, the study relied on self-reported perceptions of trust and transparency, which may introduce response bias and subjective interpretation. Participants' reported perceptions may not fully reflect actual long-term interaction behavior with music recommender systems.

Second, the study used hypothetical recommendation explanations within a controlled survey environment rather than a real-world music streaming platform. Participants

evaluated explanations in isolation, which may differ from how users experience recommendations during actual music consumption.

Additionally, the participant sample may limit the generalizability of the findings. Cultural background, music preferences, and levels of technological familiarity may influence how users interpret recommendation explanations.

Finally, the study focused only on three explanation strategies. Other explainability approaches, including conversational explanations and hybrid AI-generated explanations, were outside the scope of this study.

5.3 Implications

5.3.1 Practical Implications

This study provides practical implications for designers and developers of music recommender systems. The findings suggest that explanation design can significantly influence how users evaluate recommendation systems. Different explanation strategies produced different levels of transparency and trust, indicating that explanation type should be considered an important design component rather than a supplementary feature.

The results showed that feature-based explanations generated the highest transparency ratings, whereas lyrics-based explanations generated the highest trust ratings. Consequently, system designers should recognize that explanation strategies may support different user outcomes. If the objective is to improve users' understanding of how recommendations are generated, feature-based explanations may be more effective. If the objective is to strengthen trust in the recommender system, lyrics-based explanations may be more beneficial.

The findings also suggest that music streaming platforms may benefit from incorporating multiple forms of explanation rather than relying on a single explanation strategy. Combining information about musical features with lyric-related information may help recommendation systems support both transparency and trust simultaneously.

Overall, the results highlight the importance of carefully designing recommendation explanations. Explanation strategies should be selected based on the specific user perceptions that a recommender system aims to support, as different explanation types appear to influence transparency and trust in different ways.

5.3.2 Scientific Implications

This study contributes to explainable recommender system research by demonstrating that trust and transparency do not necessarily respond similarly to explanation strategies. The findings provide empirical evidence that explanation types influence cognitive understanding and emotional confidence differently.

The study also contributes specifically to music recommender system research by examining lyrics-based explanations, which remain relatively underexplored compared to feature-based and collaborative recommendation explanations.

Furthermore, the significant interaction effect identified through the two-way repeated-measures ANOVA strengthens the theoretical understanding of trust and transparency as distinct but related constructs within explainable AI systems.

5.4 Limitations

Several limitations should be acknowledged in this study. First, the research was conducted using survey-based evaluations rather than a fully operational music streaming environment. Users' real-world behavior and long-term trust development may differ from survey responses.

Second, the study relied on self-reported perceptions, which may be influenced by personal interpretation and response bias. Participants may evaluate explanations differently depending on their familiarity with music recommendation systems or personal attitudes toward AI technologies.

Third, the sample size and participant demographics may limit the generalizability of the findings. Cultural background, music preferences, and technological literacy may influence how users perceive recommendation explanations.

Another limitation is that the study examined only three explanation types. Other forms of explainable recommendation interfaces, including interactive explanations and conversational recommendation systems, were outside the scope of the research.

Finally, the study focused specifically on trust and transparency. Additional user-centered outcomes such as satisfaction, enjoyment, recommendation acceptance, and continued usage intention were not examined.

5.5 Recommendations for Future Research

Future research could investigate how lyrics-based explanations function within real-world streaming platforms over longer periods of interaction. Longitudinal

studies may provide deeper insight into how trust develops through repeated exposure to recommendation explanations.

Future studies could also explore hybrid explanation strategies that combine feature-based transparency with lyrics-based emotional reasoning. Such approaches may help systems optimize both cognitive understanding and emotional trust simultaneously.

Additionally, future research could investigate the influence of personal characteristics such as musical expertise, personality traits, emotional sensitivity, or trust propensity on users' responses to explanation strategies. Another promising direction would be to examine conversational and AI-generated explanations within music recommender systems. Advances in generative AI may enable recommendation systems to provide more adaptive and personalized explanation experiences.

Finally, future research could expand beyond trust and transparency by investigating how explanation strategies influence recommendation satisfaction, engagement, perceived enjoyment, and long-term adoption of music recommender systems.

6 Conclusion

As music recommender systems increasingly rely on complex algorithms, users are often given little understanding of why particular songs are recommended to them. This study explored whether different explanation strategies can influence how users perceive transparency and trust within music recommender systems. More specifically, the research examined whether lyrics-based explanations create different user perceptions compared to feature-based and example-based explanations.

The findings demonstrated that explanation type plays an important role in shaping user perceptions. Feature-based explanations produced the highest transparency scores, suggesting that explanations built around musical attributes such as genre, tempo, or audio characteristics helped users better understand the recommendation process. In contrast, lyrics-based explanations generated the highest trust scores, indicating that emotionally meaningful and semantically rich explanations based on song lyrics contributed more strongly to users' confidence in the recommender system. Example-based explanations produced comparatively weaker evaluations across both transparency and trust.

A major contribution of this study was the identification of a significant interaction effect between explanation type and outcome type. The results showed that transparency and trust do not react similarly across explanation strategies. Instead, different explanation approaches appear to influence different psychological dimensions of user perception. Feature-based explanations mainly supported cognitive understanding and system clarity, while lyrics-based explanations appeared to strengthen emotional connection and perceived credibility.The correlation and moderation analyses further supported this pattern. The relationship between transparency and trust was strongest in the lyrics-based condition, suggesting that emotionally resonant explanations may strengthen how user understanding translates into trust. In comparison, feature-based explanations, although highly transparent, demonstrated a weaker connection between transparency and trust.

These findings contribute to the growing research area of explainable recommender systems by showing that explanation design should not only focus on making systems understandable, but also on creating explanations that users perceive as meaningful and personally relevant. Within music recommendation contexts, where listening experiences are often emotional and subjective, explanations based on lyrical meaning may provide additional value beyond traditional feature-based approaches.

Taken together, the results highlight the importance of balancing cognitive transparency with emotional relevance when designing explainable music recommender systems. By demonstrating that different explanation strategies

influence transparency and trust in different ways, the study provides insights that may support the development of more human-centred and trustworthy recommendation interfaces in future music streaming platforms.

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8 Appendices

Appendix A **Questionnaires**

This appendix presents screenshots of the questionnaires used to measure participants' perceptions of transparency and trust in the experiment.

Appendix B **Prototype screenshots**

This appendix presents screenshots from the experimental music recommender system prototype used in the study.

Appendix C **Additional Statistical Analysis**

This appendix contains supplementary statistical outputs supporting the main analysis presented in Chapter 4. Correlation heatmaps are presented for each explanation condition.

7.1 Appendix A: Questionnaires

This appendix presents screenshots of the questionnaires used in the experiment to measure participants' perceptions of transparency and trust in the music recommender system. The questionnaires were presented after each explanation condition in the prototype. All items were measured using a 5-point Likert scale ranging from 1 (Strongly disagree) to 5 (Strongly agree).

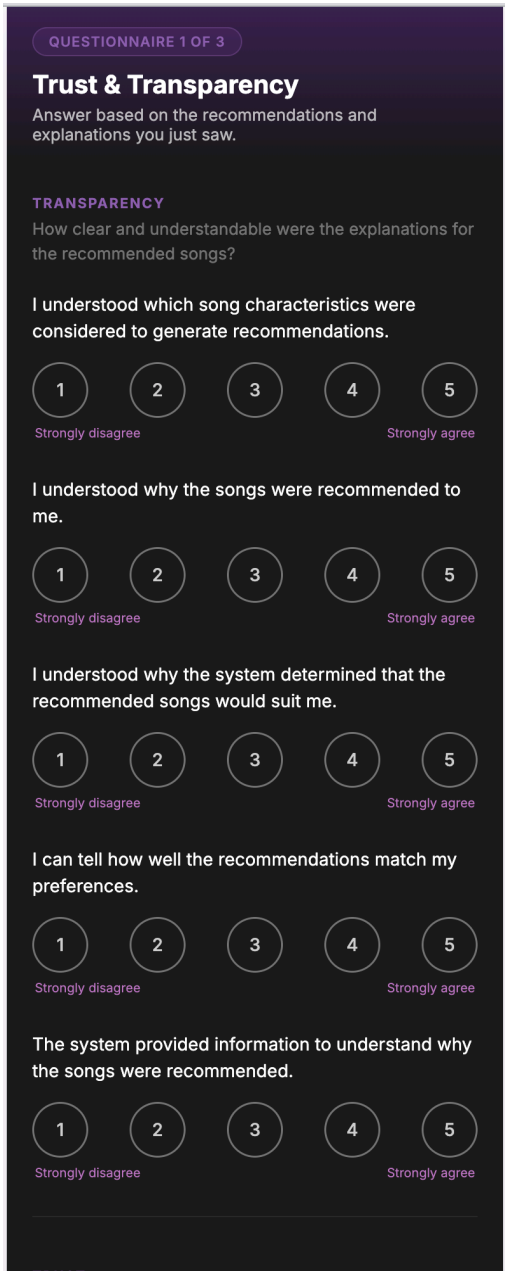


Figure A-1. Transparency questionnaire presented after each explanation condition.

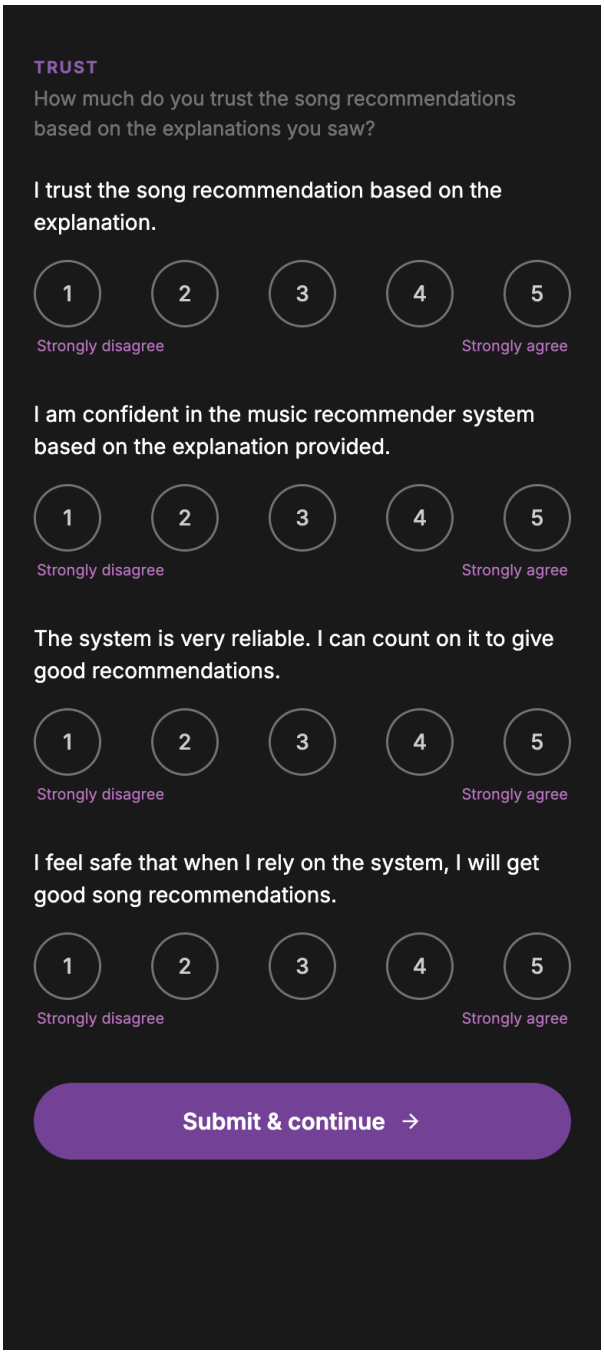


Figure A-2. Trust questionnaire presented after each explanation condition.

7.2 Appendix B: Prototype screenshots

This appendix presents screenshots from the interactive prototype used in the experiment. The prototype was developed to simulate a music recommender system in which participants interacted with different explanation types. The screenshots illustrate the song selection interface and the three explanation conditions used in the study: example-based, feature-based, and lyrics-based explanations.

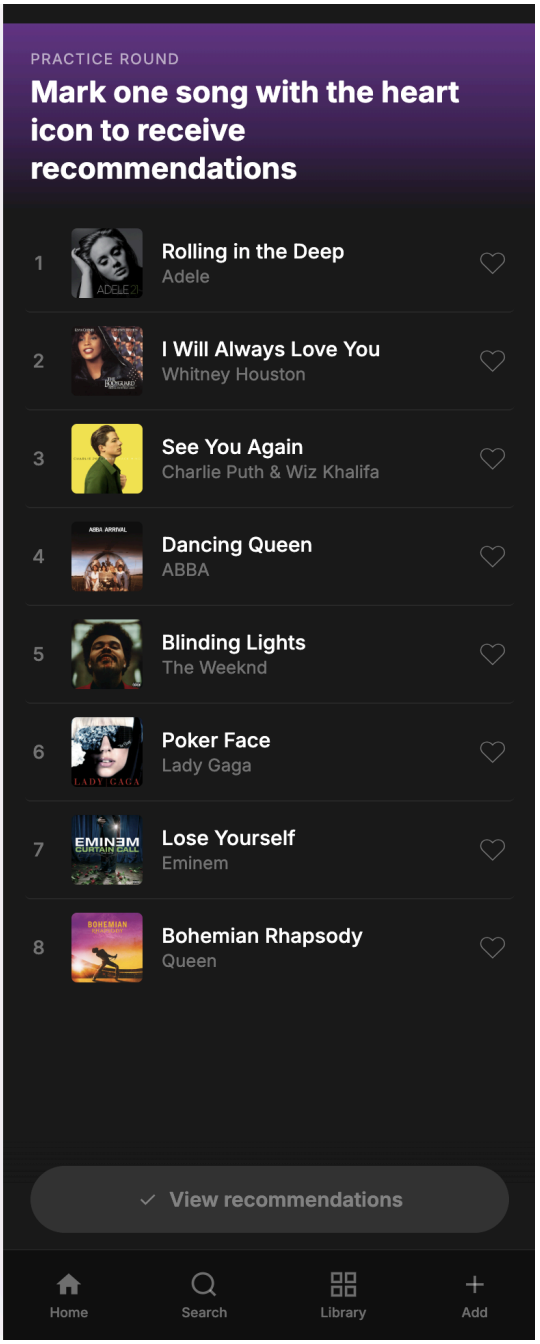


Figure B-1. Song selection interface used at the beginning of the experiment.

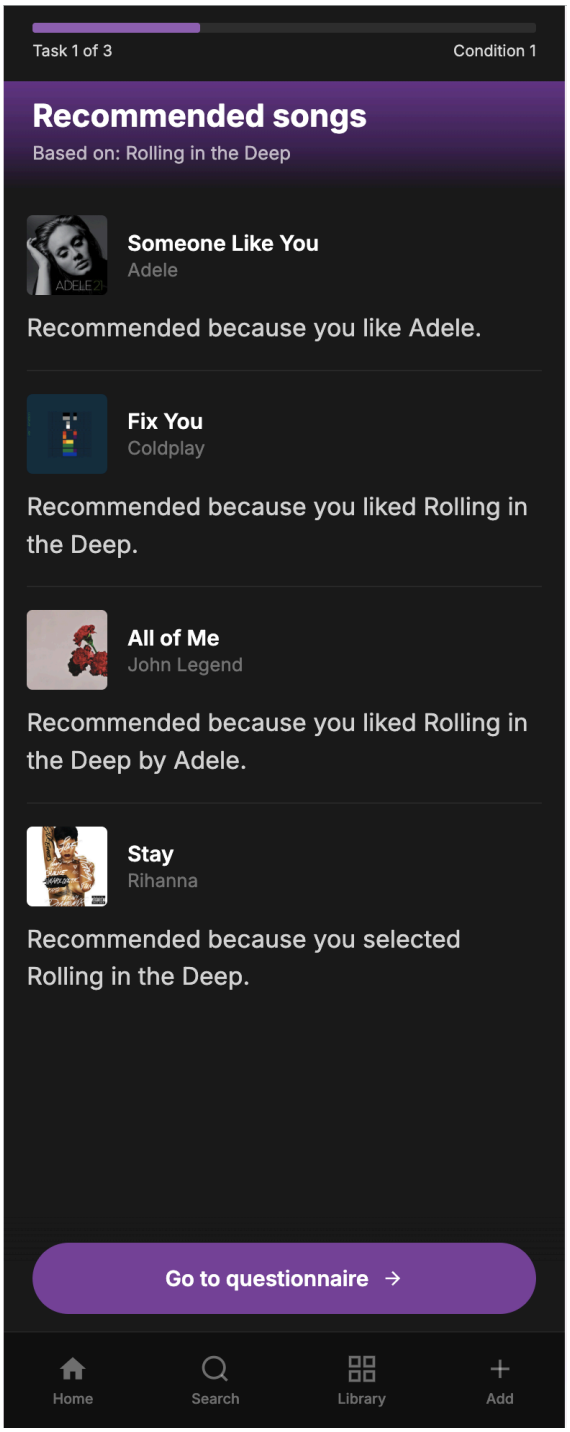


Figure B-2. Example-based explanation condition presented in the prototype.

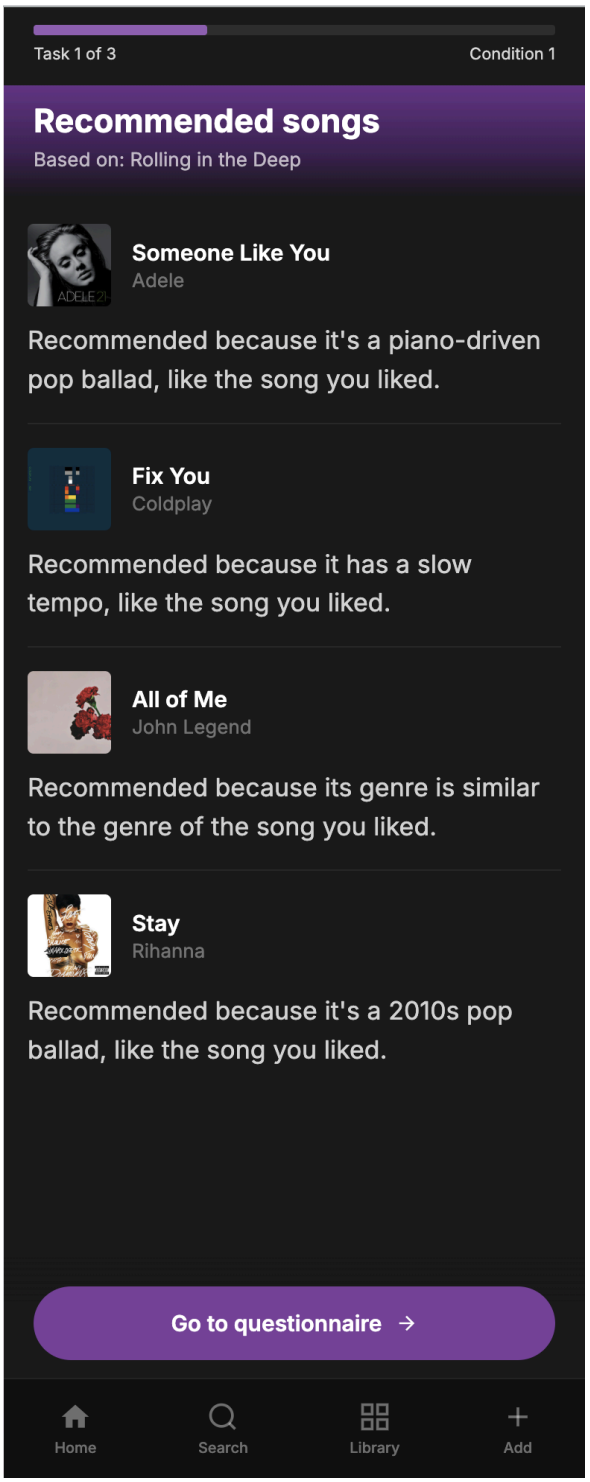


Figure B-3. Feature-based explanation condition presented in the prototype.

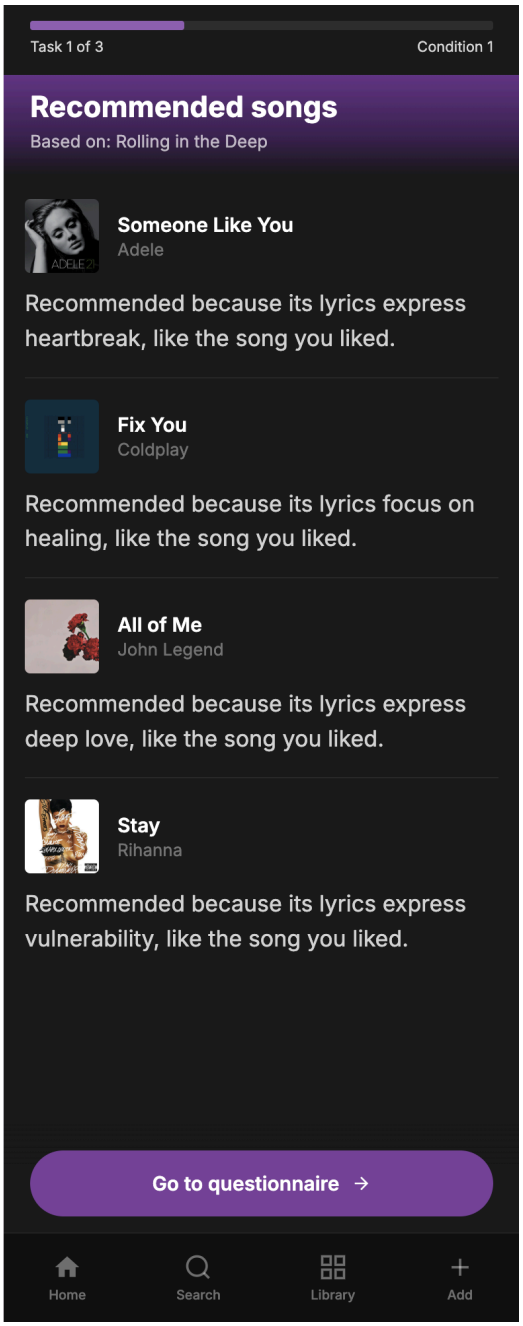


Figure B-4. Lyrics-based explanation condition presented in the prototype.

7.3 Appendix C: Additional Statistical Analysis

This appendix contains supplementary statistical outputs supporting the main analysis presented in Chapter 5.

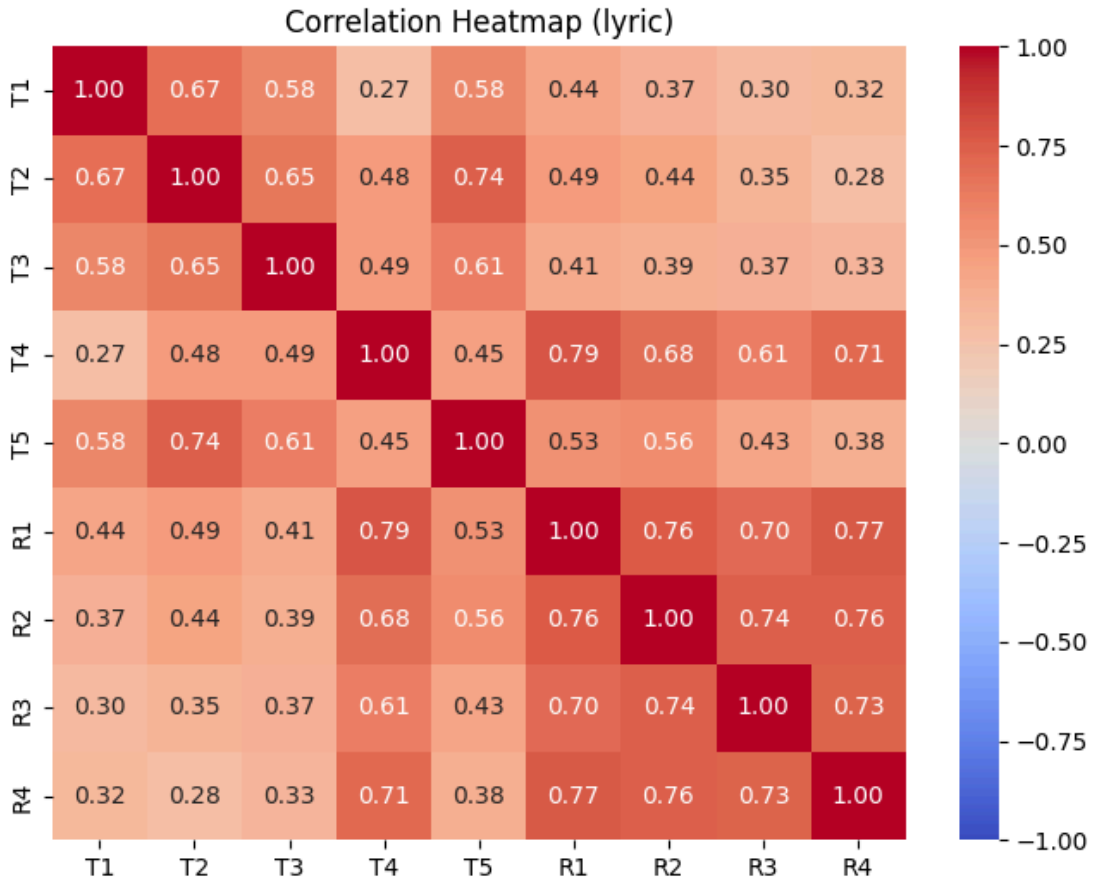


Figure C-1. Correlation heatmap for the lyrics-based explanation condition.

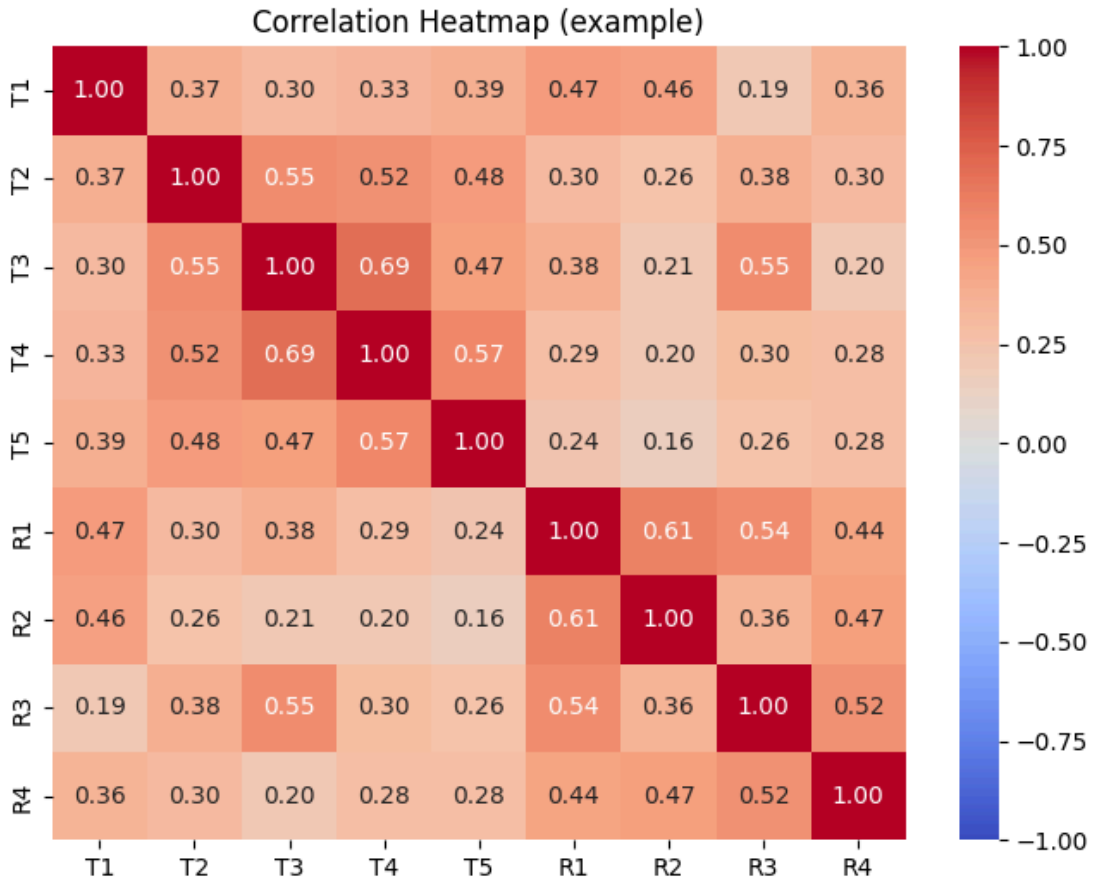


Figure C-2. Correlation heatmap for the example-based explanation condition.

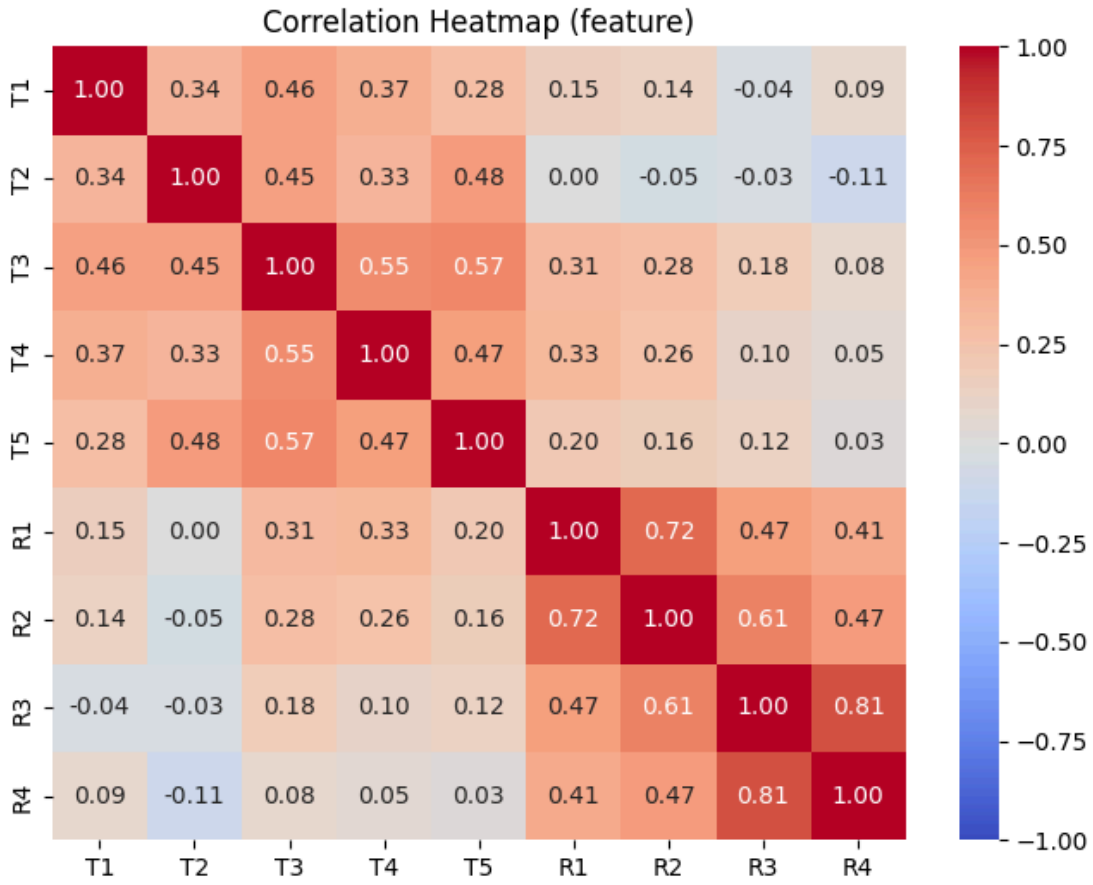


Figure C-3. Correlation heatmap for the feature-based explanation condition